Physics-Informed Neural Networks for Glucose-Insulin Dynamics

Priyal Sharma

March 2026

Contents

1 Dataset **2**
1.1 Data Source . . . 2
1.2 Dataset Structure . . . 2
1.3 Cohort-Level Summary Statistics . . . 2
1.4 Key Trajectory Statistics . . . 2

2 Data Preprocessing **2**
2.1 Feature Selection . . . 2
2.2 Cleaning . . . 2
2.3 Normalisation . . . 2
2.4 Sliding Window Construction . . . 3

3 Physics-Informed Neural Network (PINN) **3**
3.1 What is a PINN? . . . 3
3.2 Governing Equations: The Bergman Minimal Model . . . 3
3.3 Model Inputs and Outputs . . . 4
3.4 Loss Function . . . 4
3.4.1 Data Loss . . . 4
3.4.2 Physics Loss . . . 4
3.4.3 The Physics Coefficient λ . . . 5

4 Model Architectures **5**
4.1 RNN-PINN . . . 5
4.2 LSTM-PINN . . . 5
4.3 Transformer-PINN . . . 5

5 Training and Evaluation **6**
5.1 One Global Model . . . 6
5.2 Hyperparameter Grid Search . . . 6
5.3 Evaluation Metric . . . 6

1 Dataset

1.1 Data Source

The dataset was generated using `simglucose`, a Python implementation of the UVA/Padova Type 1 Diabetes Simulator (`https://github.com/jxx123/simglucose`). It simulates blood glucose dynamics for virtual patients across a variety of dietary scenarios.

1.2 Dataset Structure

File	Shape	Description
`train_dataset.csv`	(864,600 × 16)	Training trajectories
`val_dataset.csv`	(216,150 × 16)	Validation trajectories
`dataset_summary.csv`	(750 × 12)	Per-trajectory summary statistics

The dataset covers 30 virtual patients across 3 cohorts (adult, adolescent, child) and 25 dietary scenarios, yielding 750 total trajectories (600 train / 150 validation). Every trajectory spans exactly 72 hours at 3-minute resolution: 1,441 timesteps per trajectory, uniform spacing, no missing values.

1.3 Cohort-Level Summary Statistics

Cohort	BG Mean	BG Std	TIR	Hypo	Hyper	Total Insulin (U)
Adult	129.0	22.8	95.7%	1.9%	2.3%	47.5
Adolescent	137.1	31.1	88.0%	2.3%	9.7%	35.2
Child	114.3	42.2	75.5%	18.1%	6.4%	19.3

Children exhibit the highest hypoglycemia rate and the widest BG variance, reflecting greater physiological insulin sensitivity. Adults achieve the best time-in-range (TIR).

1.4 Key Trajectory Statistics

Global BG across all training rows: mean 126.97 mg/dL, std 41.98, range $[\approx 0,\ 787.68]$ (clipped to $[30, 400]$ in preprocessing). Insulin values range from 0.005 to 12.5 U/min with a mean of 0.024. Meal events (CHO > 0) occur at only 0.5% of all timesteps, with a mean of 17.55g carbohydrates per event.

2 Data Preprocessing

2.1 Feature Selection

Three input features are used: `BG` (blood glucose), `insulin`, and `CHO` (carbohydrate intake). CGM is excluded because it is a noisy, lagged proxy for BG and adds no information when true BG is available. Derived columns (LBGI, HBGI, Risk) are excluded because they are computed directly from BG.

2.2 Cleaning

BG is clipped to $[30, 400]$ mg/dL to remove physiologically implausible simulator artifacts. Insulin and CHO are zero-filled for any missing entries.

2.3 Normalisation

All three features are independently z-scored using statistics computed on the training set only:

$$\tilde{x} = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}} \tag{1}$$

The same μ and σ are applied to the validation set without refitting, preventing data leakage.

2.4 Sliding Window Construction

Input-output pairs are constructed via a sliding window over each trajectory:

- **Input window:** 60 timesteps = 3 hours of [BG, insulin, CHO], shape $(60, 3)$
- **Output horizon:** 5 timesteps = 15 minutes of BG only, shape $(5,)$

Windows never cross trajectory boundaries. Each trajectory yields $1441 - 60 - 5 = 1376$ windows, for approximately 825,600 total training samples.

3 Physics-Informed Neural Network (PINN)

3.1 What is a PINN?

A Physics-Informed Neural Network is a training methodology, not a specific architecture. A standard sequence model (RNN, LSTM, or Transformer) is regularised with an additional physics loss term that penalises violations of known governing equations. The network learns to simultaneously fit the observed data *and* satisfy the physics. Only the neural network weights and biases are learned; the physical parameters are fixed throughout training.

3.2 Governing Equations: The Bergman Minimal Model

The physical constraints come from the Bergman Minimal Model (1981), a three-compartment ODE system for glucose-insulin dynamics:

$$\frac{dG}{dt} = -(p_1 + X)\, G + p_1 G_b + D(t) \tag{2}$$

$$\frac{dX}{dt} = -p_2\, X + p_3\, I \tag{3}$$

$$\frac{dI}{dt} = -n\, I + \gamma\, \max(G - h,\ 0) + u(t) \tag{4}$$

State variables:

- G — blood glucose concentration [mg/dL]
- X — remote insulin effect [latent; never directly observed]
- I — plasma insulin concentration [U/min]

Exogenous inputs:

- $D(t)$ — glucose appearance rate from meals [mg/dL/min], derived from CHO column
- $u(t)$ — exogenous insulin delivery [U/min], taken from the insulin column

Fixed parameters (Bergman 1981; not learned during training):

Parameter	Value	Units	Meaning
p_1	0.028	min^{-1}	Glucose effectiveness
p_2	0.025	min^{-1}	Decay rate of remote insulin effect
p_3	5×10^{-5}	$(\text{mU min})^{-1}$	Insulin action on glucose uptake
n	0.090	min^{-1}	Insulin clearance rate
γ	0.006	$\text{mU L}^{-1}\,\text{min}^{-2}$	Pancreatic secretion gain
h	80.0	mg/dL	Glucose secretion threshold
G_b	80.0	mg/dL	Basal blood glucose

These are population averages derived from adult subjects. The child cohort has different insulin sensitivity, so the physics loss is most accurately specified for adults and least accurate for children.

3.3 Model Inputs and Outputs

Each model receives a normalised 60-step input window and outputs three predicted state variables over the 5-step forecast horizon:

Output	Shape	Supervised?	Reason
G_{pred}	$(B,\ 5)$	Yes	Ground truth BG available in dataset
X_{pred}	$(B,\ 5)$	No	X never observed; constrained by physics only
I_{pred}	$(B,\ 5)$	Yes	Ground truth insulin available in dataset

3.4 Loss Function

The total training loss combines a data term and a physics term:

$$\boxed{\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda \cdot \mathcal{L}_{\text{physics}}} \tag{5}$$

3.4.1 Data Loss

$$\mathcal{L}_{\text{data}} = \underbrace{\text{MSE}(G_{\text{pred}},\ G_{\text{true}})}_{\text{glucose supervision}} + \underbrace{\text{MSE}(I_{\text{pred}},\ I_{\text{true}})}_{\text{insulin supervision}} \tag{6}$$

Both terms are computed in normalised space. Glucose and insulin are weighted equally. X is excluded from $\mathcal{L}_{\text{data}}$ because it was never measured by the simulator — it is a mechanistic latent variable, not an observable quantity.

3.4.2 Physics Loss

The physics loss checks whether the predicted trajectory is consistent with the Bergman ODEs (Equations 2–4). Time derivatives over the forecast horizon are approximated by forward finite differences at $\Delta t = 3$ min:

$$\left.\frac{dG}{dt}\right|_t \approx \frac{G_{t+1} - G_t}{\Delta t} \tag{7}$$

The ODE residuals at each interior timestep t of the horizon are (using Equations 2–4):

$$r_G^{(t)} = \frac{G_{t+1} - G_t}{\Delta t} - \Big[-(p_1 + X_t)\, G_t + p_1\, G_b + D_t\Big] \tag{8}$$

$$r_X^{(t)} = \frac{X_{t+1} - X_t}{\Delta t} - \Big[-p_2\, X_t + p_3\, I_t\Big] \tag{9}$$

$$r_I^{(t)} = \frac{I_{t+1} - I_t}{\Delta t} - \Big[-n\, I_t + \gamma\, \max(G_t - h,\ 0) + u_t\Big] \tag{10}$$

A residual of zero means the predicted state transition is exactly consistent with the Bergman model. The physics loss aggregates all three residuals across the $H - 1 = 4$ interior horizon steps:

$$\mathcal{L}_{\text{physics}} = \frac{1}{(H-1)\cdot 3} \sum_{t=1}^{H-1} \Big[\big(r_G^{(t)}\big)^2 + \big(r_X^{(t)}\big)^2 + \big(r_I^{(t)}\big)^2\Big] \tag{11}$$

3.4.3 The Physics Coefficient λ

λ in Equation 5 controls how strongly the physics is enforced relative to the data:

λ	**Effect**
0.0	Physics loss disabled; reduces to a pure data-driven baseline
0.001	Very light physics regularisation
0.01	Moderate physics enforcement
0.1	Strong physics enforcement
1.0	Physics dominates; may overfit to the fixed Bergman parameters

We grid search over $\lambda \in \{0.0,\ 0.001,\ 0.01,\ 0.1,\ 1.0\}$ for each architecture independently.

4 Model Architectures

All three architectures share the same input shape $(B, 60, 3)$, output structure $(B, 5, 3)$ for $[G, X, I]$, physics loss module (Equations 8–11), optimizer (Adam, $lr = 10^{-3}$), and learning rate schedule (cosine annealing, $\eta_{\min} = 10^{-5}$). Gradient norms are clipped to 1.0. The only difference between models is the sequence encoder.

4.1 RNN-PINN

The vanilla RNN updates a single hidden state at each timestep:

$$h_t = \tanh(W_{hh}\, h_{t-1} + W_{xh}\, x_t + b) \tag{12}$$

After processing all 60 input steps, the final hidden state h_{60} is decoded to (G, X, I) predictions for all five horizon steps simultaneously. **Configuration:** 2 stacked layers, hidden size 128.

The RNN's primary limitation is vanishing gradients: gradients from early timesteps shrink through 60 successive tanh operations. It is included as a lower bound to isolate the contribution of the LSTM's gating mechanism.

4.2 LSTM-PINN

The LSTM adds three multiplicative gates and a separate cell state c_t to the RNN:

$$f_t = \sigma(W_f\, [h_{t-1},\, x_t] + b_f) \quad \text{(forget gate)} \tag{13}$$
$$i_t = \sigma(W_i\, [h_{t-1},\, x_t] + b_i) \quad \text{(input gate)} \tag{14}$$
$$o_t = \sigma(W_o\, [h_{t-1},\, x_t] + b_o) \quad \text{(output gate)} \tag{15}$$
$$c_t = f_t \odot c_{t-1} + i_t \odot \tanh(W_c\, [h_{t-1},\, x_t] + b_c) \tag{16}$$
$$h_t = o_t \odot \tanh(c_t) \tag{17}$$

The cell state acts as a gradient highway, allowing the LSTM to retain context across the full 3-hour window without vanishing gradients. This is especially important for glucose dynamics, where basal insulin effects are slow and persistent (hours) while meal spikes are sharp and transient. **Configuration:** 2 stacked layers, hidden size 128. Same two-layer decoder as the RNN.

4.3 Transformer-PINN

The Transformer processes all 60 input timesteps simultaneously via multi-head self-attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \tag{18}$$

Because self-attention is permutation-invariant, sinusoidal positional encodings are added to the projected input embeddings before attention is applied:

$$\mathrm{PE}(\mathrm{pos},\, 2i) = \sin\Big(\frac{\mathrm{pos}}{10000^{2i/d_{\mathrm{model}}}}\Big) \tag{19}$$

$$\mathrm{PE}(\mathrm{pos},\, 2i+1) = \cos\Big(\frac{\mathrm{pos}}{10000^{2i/d_{\mathrm{model}}}}\Big) \tag{20}$$

The encoder output is mean-pooled over the sequence dimension to produce a global context vector, which is then decoded to (G, X, I) predictions. **Configuration:** $d_{\mathrm{model}} = 64$, 4 attention heads, 2 encoder layers, feedforward dimension 256.

Unlike RNN and LSTM, the Transformer has no sequential inductive bias and no vanishing gradient problem. It can attend directly to any timestep in the input window regardless of distance, making it well-suited to capturing long-range patterns such as the relationship between a meal bolus and the glucose peak 60–90 minutes later.

5 Training and Evaluation

5.1 One Global Model

A single model is trained on all 30 patients across all 3 cohorts. The justification is that the Bergman model structure is assumed to be the same across cohorts, with only the parameter magnitudes differing between adults, adolescents, and children, and the neural network compensating for inter-cohort variation through its learned weights. Results are reported both in aggregate and broken down per cohort to evaluate this assumption.

5.2 Hyperparameter Grid Search

For each of the three architectures, we grid search over $\lambda \in \{0.0, 0.001, 0.01, 0.1, 1.0\}$, yielding 15 total training runs. All other hyperparameters are held fixed across the grid so that only λ varies between runs.

5.3 Evaluation Metric

The primary metric is Root Mean Squared Error (RMSE) computed in physical units (mg/dL) after inverse-transforming the normalised predictions:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\big(G_{\mathrm{pred},i} - G_{\mathrm{true},i}\big)^2} \tag{21}$$

RMSE is evaluated on the 15-minute (5-step) forecast horizon across all validation trajectories.